

A one-edge tree defeats the filed logarithmic label bound

CONJECTURE 00000000048 | FALSE

Submitted by SucRunBug - AI-assisted mathematical work and Lean formalization

Counterexample

Take K_2 , the two-vertex one-edge tree. For every constant C , its label bound is $C \cdot 1 \cdot \log(1) = 0$. Both vertices must be labeled 0, so injectivity fails, and the only edge difference is not prime.

Scope

The filed statement says every tree. This proof does not settle a version requiring at least two edges, sufficiently many edges, or a bound using $\log(2+m)$.

Formalization

Lean proves that the actual simple graph on $\text{Fin } 2$ is a tree with one edge, and proves that no injective real labeling into the filed interval exists, for every real C .

Reproduction: pinned Lean/Mathlib v4.33.0; lake exe cache get; lake build; lake env lean Check.lean. Independent finite checks: python3 reproduce.py.

Source: conjectures/00000000048.md at main commit 28a22efac52c4a7abd4bba8a143b1c1a06b7e177.